RUTGERS Department of Statistics

Polynomial Scenario Complexity for Value Estimation in Multistage Stochastic Programming
— Breaking the Curse of Dimensionality

Yuchen Mao

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JOINT WORK WITH



Yifan Hu
Rutgers University

1. **Background**

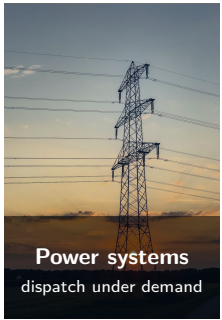
2. Illustration with Three-Stage Problems

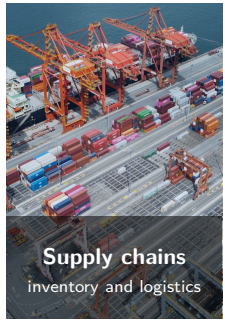
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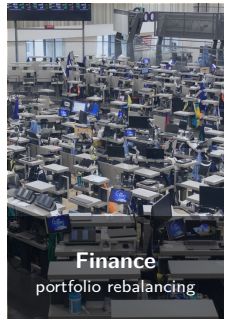
4. Takeaways and Future Directions

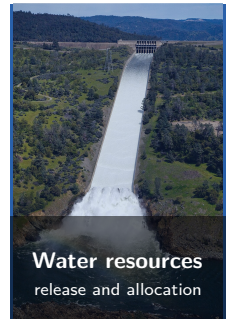
Multistage stochastic programming (MSP)

plans a sequence of decisions while uncertainty unfolds over multiple periods









The Nested Structure of MSP

Stage 3 · observe ξ_3, decide x_3

$$Q_3(x_2, \xi_3) = \inf_{x_3 \in \mathcal{X}_3(x_2, \xi_3)} f_3(x_3, \xi_3)$$

$\mathbb{E}_2[\,\cdot\,]$

Stage 2 · observe ξ_2, decide x_2

$$Q_2(x_1, \xi_2) = \inf_{x_2 \in \mathcal{X}_2(x_1, \xi_2)} \{ f_2(x_2, \xi_2) + \mathbb{E}_2[Q_3(x_2, \xi_3)] \}$$

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Stage 1 · evaluate x_1

$$F(x_1) = f_1(x_1, \xi_1) + \mathbb{E}_1[Q_2(x_1, \xi_2)]$$

- Each $\mathbb{E}_t[\,\cdot\,]$ is a **conditional expectation given the observed history $\xi_{[t]} = (\xi_1, \ldots, \xi_t)$**, generally not available in closed form
- The general recursion repeats the same pattern and we use $T = 3$ throughout the talk

$$Q_t(x_{t-1}, \xi_t) = \inf_{x_t \in \mathcal{X}_t(x_{t-1}, \xi_t)} \{ f_t(x_t, \xi_t) + \mathbb{E}_t[Q_{t+1}(x_t, \xi_{t+1})] \}$$

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Estimate the value of one prescribed first-stage decision

$$F(x_1) = \underbrace{f_1(x_1, \xi_1)}_{\text{known}} + \underbrace{\mathbb{E}_1[Q_2(x_1, \xi_2)]}_{\text{to be estimated}}$$

Sampling oracle

Conditional draws of ξ_{t+1} given $\xi_{[t]}$; stagewise dependence allowed

Goal

Root mean squared error at most ϵ

Complexity

Expected number of scenario paths (*scenario complexity*)

This problem is believed to inherit the curse of dimensionality

Scenario paths grow exponentially in the number of stages T (Shapiro, 2006; Lan and Zhou, 2021)

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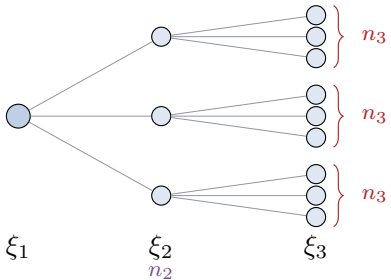
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$$\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1) \coloneqq f_1(x_1,\xi_1) + \frac{1}{n_2}\sum_{i=1}^{n_2} \inf_{x_2 \in \mathcal{X}_2(x_1,\xi_2^{(i)})} \left\{ f_2(x_2,\xi_2^{(i)}) + \frac{1}{n_3}\sum_{j=1}^{n_3} \inf_{x_3 \in \mathcal{X}_3(x_2,\xi_3^{(i,j)})} f_3(x_3,\xi_3^{(i,j)}) \right\}$$

General dependence

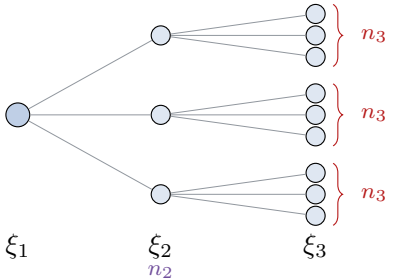


With **general dependence** the estimator runs over $n_2 \times n_3$ **scenario paths**

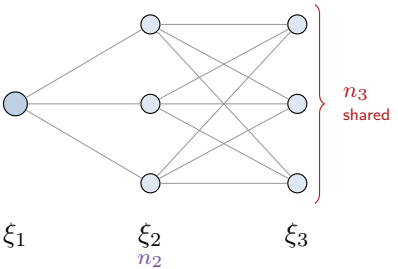
Nested SAA: the Benchmark

$$\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1) \coloneqq f_1(x_1,\xi_1) + \frac{1}{n_2}\sum_{i=1}^{n_2} \inf_{x_2 \in \mathcal{X}_2(x_1,\xi_2^{(i)})} \left\{ f_2(x_2,\xi_2^{(i)}) + \frac{1}{n_3}\sum_{j=1}^{n_3} \inf_{x_3 \in \mathcal{X}_3(x_2,\xi_3^{(i,j)})} f_3(x_3,\xi_3^{(i,j)}) \right\}$$

General dependence



Stagewise independence



With **general dependence** the estimator runs over $n_2 \times n_3$ **scenario paths**

Under stagewise independence, samples can be shared across nodes, enabling algorithms such as stochastic dual dynamic programming (SDDP) (Pereira and Pinto 1991)

Mean squared error Decomposition

$$\mathbb{E}_1\left[|\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1) - F(x_1)|^2\right] = \left|\mathbb{E}_1[\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1)] - F(x_1)\right|^2 + \mathrm{Var}_1\left(\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1)\right)$$

$$= \underbrace{\left|\mathbb{E}_1\left[\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1) - \hat{F}_{\mathrm{SAA}}^{n_2,\infty}(x_1)\right]\right|^2}_{\text{bias, controlled by } n_3} + \underbrace{\mathrm{Var}_1\left(\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1)\right)}_{\text{variance, controlled by } n_2}$$

where we use $\mathbb{E}_1[\hat{F}_{\mathrm{SAA}}^{n_2,\infty}(x_1)] = F(x_1)$, the $n_3 \to \infty$ estimator is unbiased

Bias

The infimum acts on a noisy inner average:

$$0 \le F(x_1) - \mathbb{E}_1\left[\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1)\right] = \mathcal{O}\left(n_3^{-1/2}\right)$$

$$\Longrightarrow \quad n_3 \asymp \epsilon^{-2}$$

Variance

The outer average of n_2 i.i.d. copies fluctuates:

$$\mathrm{Var}_1\left(\hat{F}_{\mathrm{SAA}}^{n_2,n_3}(x_1)\right) = \mathcal{O}\left(n_2^{-1}\right)$$

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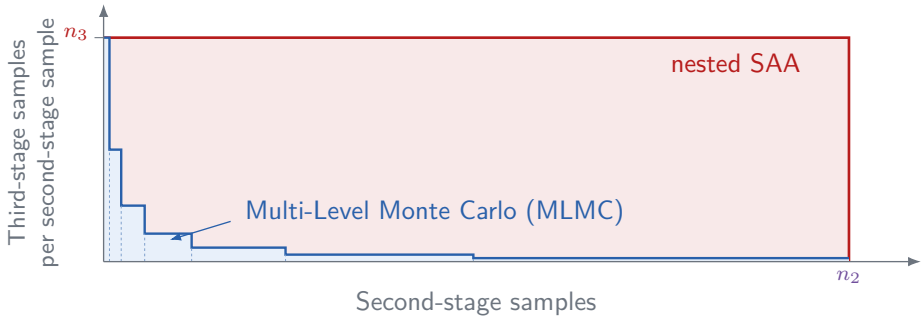
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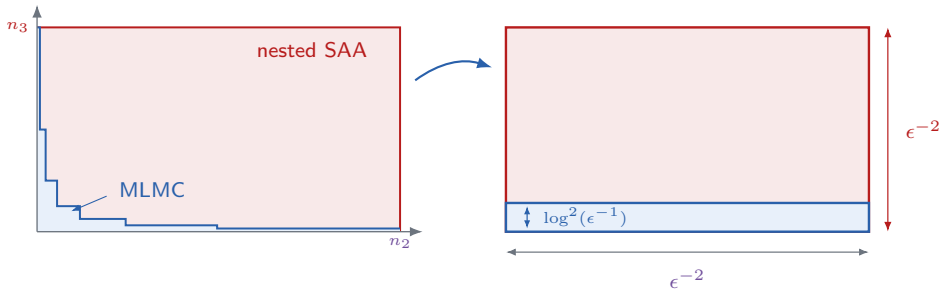
$$\underbrace{\mathcal{O}(\epsilon^{-4})}_{\text{nested SAA}} \longrightarrow \underbrace{\mathcal{O}(\epsilon^{-2}\log^2(\epsilon^{-1}))}_{\text{MLMC}}$$



*The additional tail is omitted because its area is negligible.

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The following construction is based on one given second-stage sample ξ_2

- **The level-ℓ sample value function:** for $\ell \in \{0, 1, \ldots, M_2\}$

$$\hat{h}_2^{\ell}(x_1, \xi_2) := \inf_{x_2 \in \mathcal{X}_2(x_1, \xi_2)} \left\{ f_2(x_2, \xi_2) + 2^{-\ell} \sum_{j=1}^{2^{\ell}} Q_3(x_2, \xi_3^{(j)}) \right\}$$

  - scenario paths: 2^{ℓ} per second-stage sample
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- **Match the bias of nested SAA:** choose the finest level M_2 so that

$$2^{M_2} \asymp n_3 \implies M_2 = \mathcal{O}(\log(\epsilon^{-1})) \implies \text{finest-level bias } \mathcal{O}(\epsilon)$$

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Correction Construction: Sample Reuse to Control the Variance

- **Correction requirements:**
  - Mean condition:
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    a small $q_2(\ell)$ in $\hat{H}_2 = \Delta_2^{\lambda_2}/q_2(\lambda_2)$ can otherwise amplify the second moment

- **Reuse the same block:** split the 2^ℓ samples into odd and even halves. Each half has the same conditional distribution as $\hat{h}_2^{\ell-1}$

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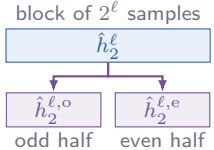
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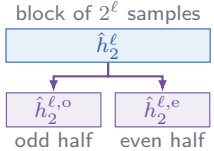
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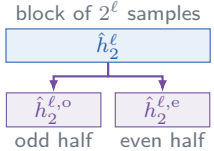
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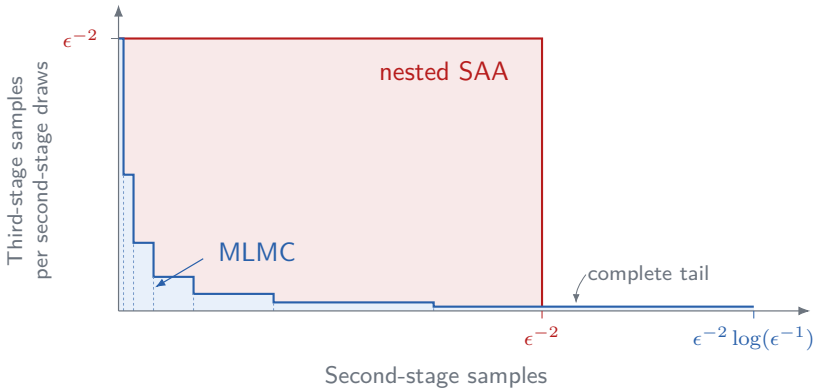
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Scenario Complexity: Nested SAA versus MLMC



1. Background

2. Illustration with Three-Stage Problems

3. **Extension to Multistage Problems**

4. Takeaways and Future Directions

- **Recall the three-stage construction:** sample value functions across levels, level sampling, sample reuse, outer average
- **Backward recursion** at stages $t = T-1, \dots, 2$: the inner average runs over the recursive $\hat{H}_{t+1}(x_t, \xi_{t+1})$ instead of the exact $Q_{t+1}(x_t, \xi_{t+1})$

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$$\hat{H}_t(x_{t-1}, \xi_t) := \frac{\Delta_t^{\lambda_t}(x_{t-1}, \xi_t)}{q_t(\lambda_t)}$$

- **Main barrier to recursion:** $\Delta_t^{\lambda_t}$ is generally a difference of convex value estimators; placing it inside the preceding-stage minimization may not preserve convexity

  For $T = 3$, the large stage-2 sample makes the convex component dominate the stage-3 correction, preserving convexity.

Nested SAA (Shapiro, 2006)

$$\underbrace{n_2}_{\epsilon^{-2}} \times \underbrace{n_3}_{\epsilon^{-2}} \times \cdots \times \underbrace{n_T}_{\epsilon^{-2}} = \mathcal{O}\left(\epsilon^{-2(T-1)}\right)$$

bias control sets $n_3, \ldots, n_T$; variance control sets n_2; each stage contributes ϵ^{-2}

MLMC value estimator

$$\underbrace{n_2}_{\epsilon^{-2}\log^{2T-5}(\epsilon^{-1})} \times \underbrace{\mathcal{O}(M_2)}_{\log(\epsilon^{-1})} \times \cdots \times \underbrace{\mathcal{O}(M_{T-1})}_{\log(\epsilon^{-1})} = \mathcal{O}\left(\epsilon^{-2}\log^{3T-7}(\epsilon^{-1})\right)$$

bias control at stages $2, \ldots, T-1$ now spends only $\log(\epsilon^{-1})$ each; the saving inflates the variance, so n_2 carries the extra log factors

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1. Background
2. Illustration with Three-Stage Problems
3. Extension to Multistage Problems
4. **Takeaways and Future Directions**

- **Nested SAA branches at every conditional layer:** $\mathcal{O}(\epsilon^{-2(T-1)})$
- **Recursive MLMC keeps the ϵ^{-2} leading power:** $\mathcal{O}(\epsilon^{-2}\log^{3T-7}(\epsilon^{-1}))$
- **To our knowledge, the first estimator breaking the curse of dimensionality for multistage value estimation under general dependence**

Is multistage stochastic programming as difficult as nested SAA suggests? Probably not

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- **Move signed differences outside the stagewise infima**
- **Combine adaptive level sampling with first-order methods**

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Not yet, much remains to be explored

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Thank You

QUESTIONS WELCOME

PERSONAL WEBSITE



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